1) Delete the first node

```
if (head == NULL) // check that list is empty
struct Node* temp = head; // store first node
```

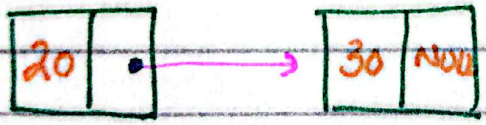
head → 10
temp → 10

```
head = head->next;        // move head to next node
```

temp → 10 (old first node)

head → 20 → 30 → NULL

```
free (temp); // free old first node
```



a) Delete Last Node

```
if (head == NULL) // check, list is empty
if (head->next == NULL) // check, there is only one node
struct Node* temp = head;
                  // initialize temp make it equal to head
```

temp = 10

head → 10

temp → 10

check node after the node exists

```
while (temp -> next -> next != NULL)
{   temp = temp->next;  }
```

itr 1

10 → 20 → 30

30 =! Null

temp = 20

itr 2

20 → 30 → 40

40 =! NULL

temp = 30

itr 3

30 → 40 → NULL

Condition is false

loop stops.

temp → 30 second last node

temp → 40 last node

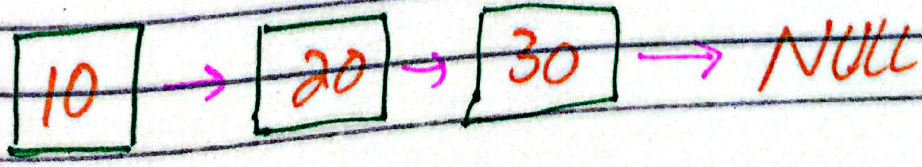
```
free ( temp -> next)
```

deleting last node

temp = 40 , deleted

```
temp -> next = NULL;
```

Node 30 points to null



3| Delete node at given Position.

Let Position = 3

```
if (head == NULL)
```
check list is empty

```
if (pos < 1)
```
check position is valid
3 < 1 false

```
if (pos == 1)
```
check position is 1
3 == 1 false

```
Struct Node* temp = head
```
intialize temp.
temp = head, temp = 10

```
for int i=1 , i < pos-1 && temp != NULL ; i++)
    { temp = temp -> next ; }
```

itr 1 i=1 1<2 temp = 20
itr 2 i=2 2<2 Condition false

temp → 20 node before position 3

```
if (temp == NULL || temp -> next == NULL)
```

temp → 20
temp -> next → 30

Neither = NULL
Condition False

```
struct Node* del = temp -> next
```
Store node to delete

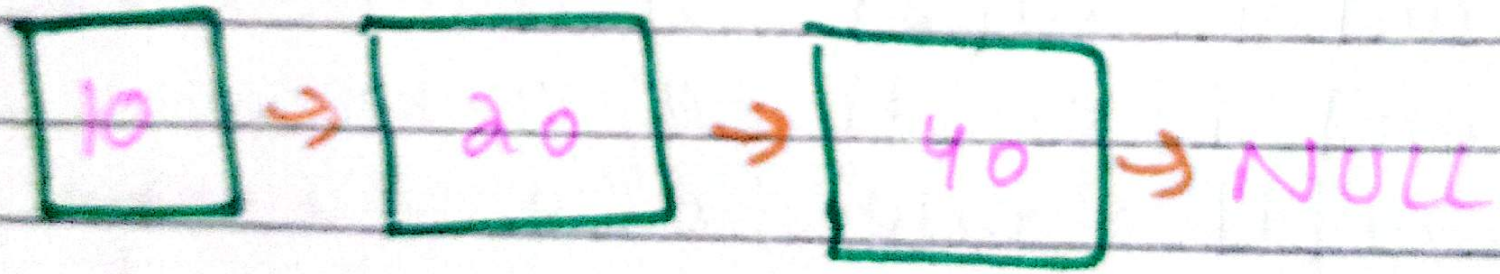
temp → 20
del → 30

```
temp -> next = del -> next;
```

Connecting Node 20 to 40

```
free ( del);
```

freeing the deleted node



CamScanner